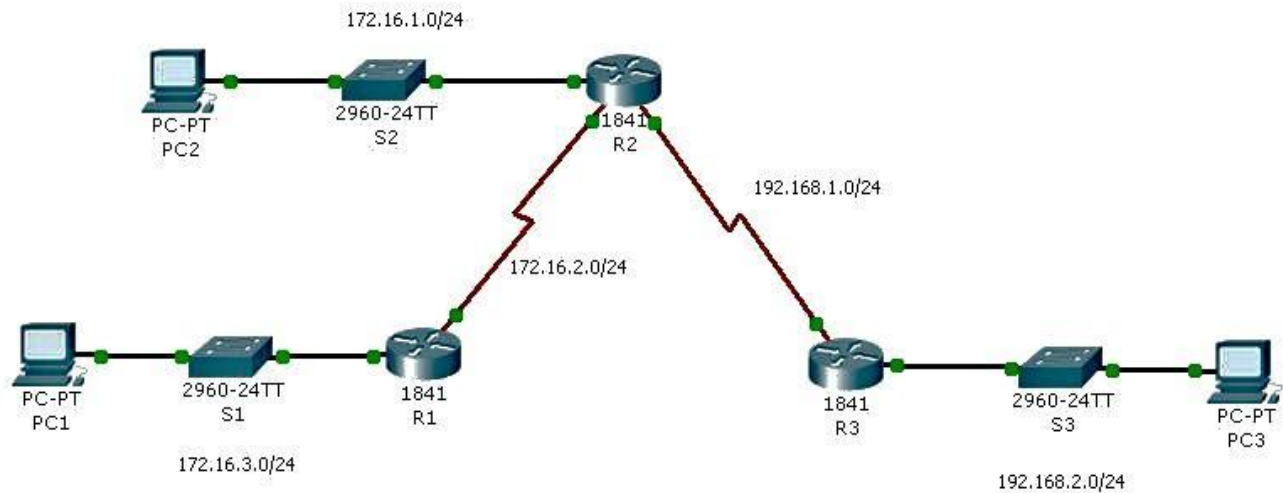




Lab 5 – Dynamic Routing RIP v2

The goal of this Lab is to configure the dynamic routing RIP v2 in a network.

Consider the following topology:



We have 3 LAN attached to 3 routers. The purpose of our lab will be to interconnect different LANs to communicate in the network.

Addressing the 3 networks:

- 172.16.1.0/24
- 172.16.3.0/24
- 192.168.1.0/24

With respect to IP addressing between routers, we have:

- 172.16.2.0/24 which represents the network address between Router 1 and Router 2.
- 192.168.2.0/24 between the 2nd and the 3rd.

1. Creating the topology

Please add the proper interface to use the serial cables.

2. Configuring IP Information on Ethernet Interfaces on a Network

Configure the following IP information:

Peripheral	Interface	IP Address	Subnetwork Mask	Gateway
R1	Fa0/0	172.16.3.1	255.255.255.0	N.A.
	S0/0/0	172.16.2.1	255.255.255.0	N.A.
R2	Fa0/0	172.16.1.1	255.255.255.0	N.A.

	S0/0/0	172.16.2.2	255.255.255.0	N.A.
	S0/0/1	192.168.1.2	255.255.255.0	N.A.
R3	Fa0/0	192.168.2.1	255.255.255.0	N.A.
	S0/0/1	192.168.1.1	255.255.255.0	N.A.
PC1		172.16.3.10	255.255.255.0	172.16.3.1
PC2		172.16.1.10	255.255.255.0	172.16.1.1
PC3		192.168.2.10	255.255.255.0	192.168.2.1

3. Verification of the state of the interfaces

Check the state of the interfaces on the routers by entering the command **show ip interface brief**.

4. Checking the information on the routing table

Check the contents of the IP routing table by entering the **show ip route** command on all three routers. The IP routing table illustrates the directly connected networks, as well as the associated interface.

5. Configuration of the routing protocol

- Access router 1 and activate the config mode.
- In order to configure the RIP dynamic routing protocol, simply type the **router rip** command.
- Note: To see the different routing protocols, you type the **router** command, followed by a **?**.
- To configure the RIP V2, you must specify the version. To do this, type the **version 2** command.
Note: **?** allows to see the commands at a given level of the configuration, do not hesitate to use it.
- After choosing RIP V2, you must list the networks directly connected to this router by typing the **network** command followed by the IP address of the network, example: **network 172.16.3.0**.
- The same procedure is repeated on the other routers.
- Check the routing tables and note the routes added by the RIP v2 protocol.

6. Test

Do some tests and make sure all the machines are connected.